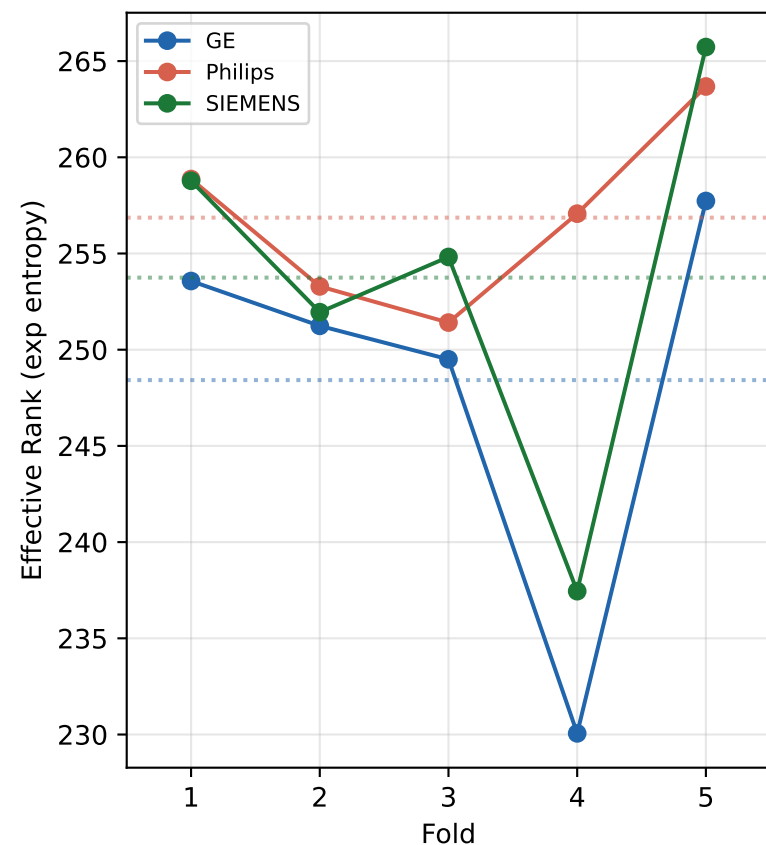
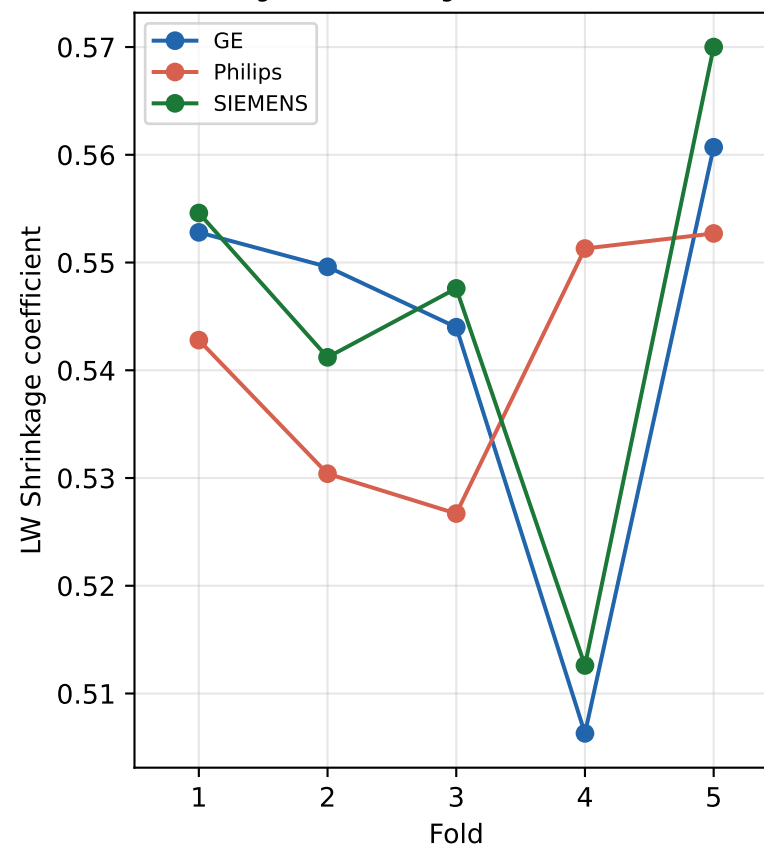


# Covariance Geometry by Manufacturer — Fold-Local Analysis (LW covariance fitted on trainDev, applied to trainDev subgroups)

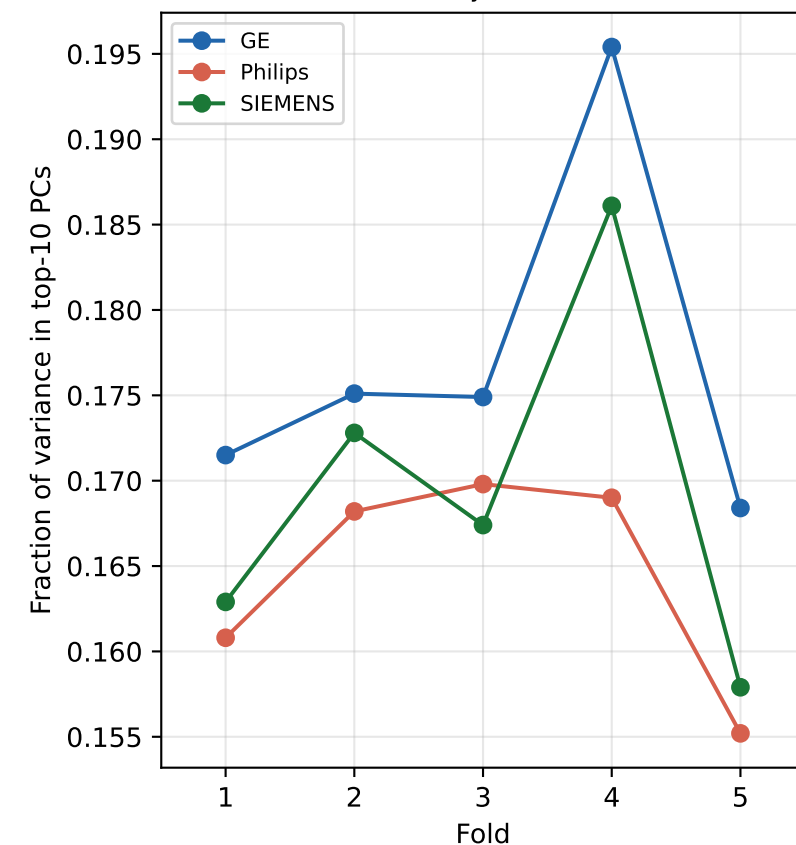
### Effective Rank per Manufacturer (LW covariance, trainDev fitted)



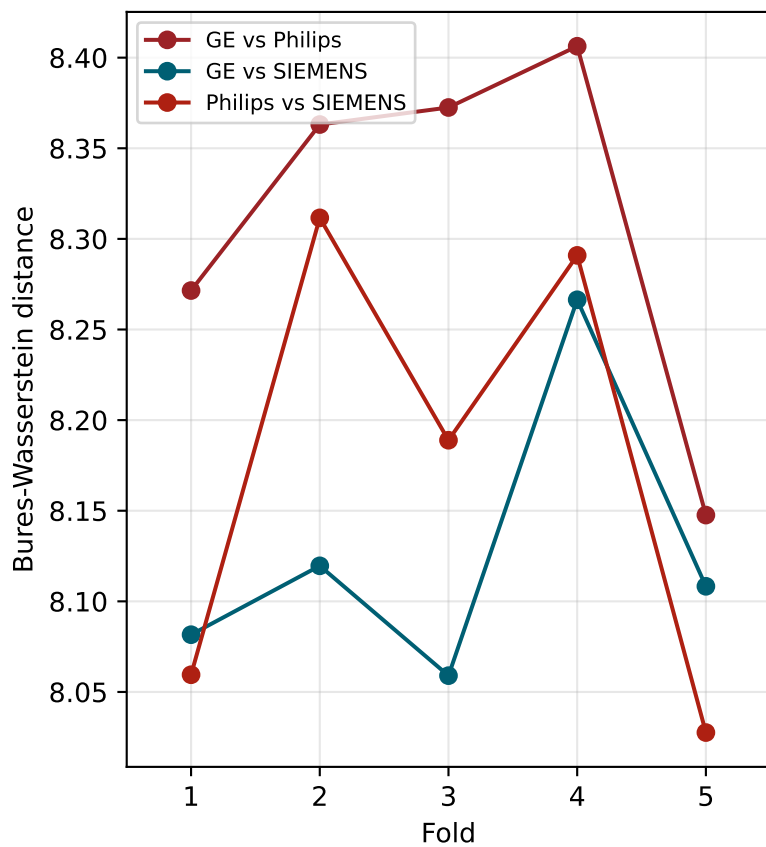
### Ledoit-Wolf Shrinkage (higher = more regularization needed)



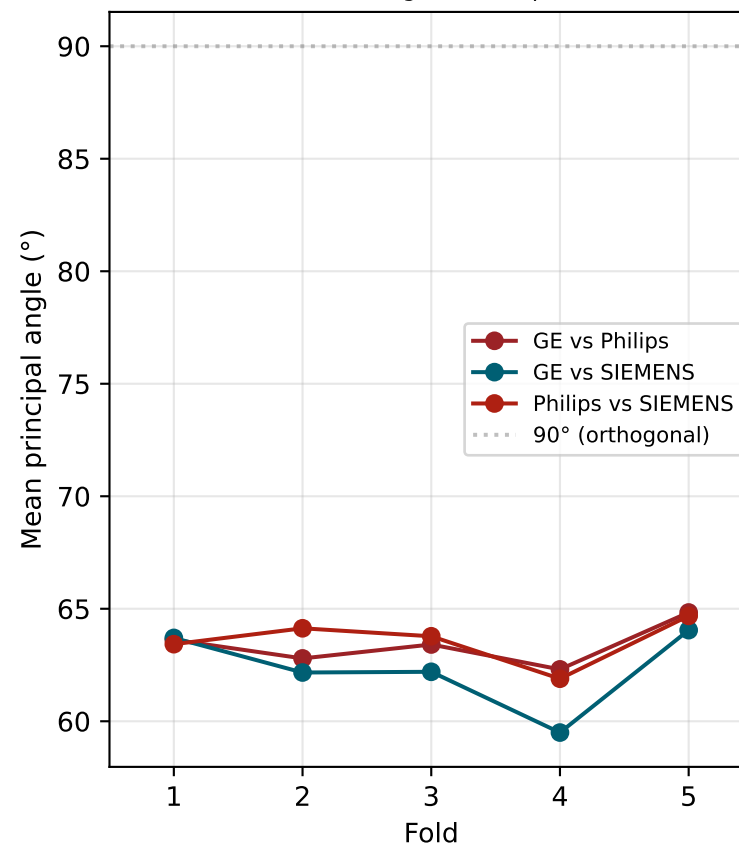
### Top-10 PC Variance Fraction (dimensionality concentration)



### Bures-Wasserstein Distance (between manufacturer covariances)



### Mean Principal Angle (top-20 PCs) (90° = orthogonal subspaces)



### kNN-5 Manufacturer BACC vs permutation null

